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AI Enabled Autonomous Gas-lift Optimization: Challenges and Innovations from Digital Oil Field in Abu Dhabi

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Abstract

This paper presents the deployment of an AI-enabled autonomous gas-lift optimization system, integrating real-time centralized Advanced Process Control (APC) with cloud-based analytics to enhance artificial gas lift performance in producer wells. The objective is to assess system performance, identify operational and technical challenges, and highlight innovative practices in overcoming these barriers. The paper emphasizes work process transformation, legacy system integration, and cross-functional collaboration to deliver a scalable, adaptive, and production-optimized control solution in complex field environments.

Deployments across two hundred wells demonstrated up to 2% increase in oil production and up to 5% - 30% optimization in lift gas consumption at a well level, while maintaining alignment with facility integrity limits and operational constraints. Flowing bottomhole pressure compliance within the optimal range 100% of the time. These outcomes were achieved through coordinated workflows among field operators, control engineers, and petroleum engineers, using automated processes deployed both in the field and the cloud.

Field operational work processes included sensor validation, real-time injection adjustments, and system override logging. Cloud-hosted workflows supported data ingestion, automated model retraining, well-specific requirements, and asynchronous deployment of updated operating envelopes for continuous optimization.

Several challenges were encountered and addressed: sensor and valve failures necessitated the creation of diagnostics and fallback logic; low-fidelity well models were improved using AI-driven recalibration based on field testing; integration with legacy DCS systems required standardized interfaces; cybersecurity and OT-IT data bridging were managed with secure protocols. Limited historical data and unavailability of measurements from wells reduced AI model accuracy, which was mitigated using inferential models, synthetic data generation, and transfer learning. These innovations enabled scalable, autonomous gas-lift optimization while maintaining system reliability and adaptability across varied field conditions.

Introduction

Artificial lift systems play a crucial role in maintaining production rates in mature oil fields. Among these, gas lift system is widely used due to its flexibility and applicability across a wide range of reservoirs and well conditions. Traditionally, gas-lift optimization has relied on manual analysis and rule-based control schemes that are both time-consuming and reactive in nature. With the growing complexity of field operations and the volume of real-time data, manual approaches fall short in achieving continuous, optimized performance.

Autonomous optimization using AI-enabled control systems offers a paradigm shift. With access to continuous data streams and computational tools, it is now possible to achieve real-time optimization of gas-lifted wells. This Autonomous Control system exemplifies this shift by introducing a scalable, adaptive, and intelligent system that operates well autonomously, adhering to operational and safety constraints. This study builds upon the foundational work presented in Hernandez et al (2024), where a Multivariable Predictive Control (MPC) framework was successfully deployed to enable autonomous well control in natural flow and gas lift wells. That implementation demonstrated how physics-based models combined with AI algorithms can dynamically manage multiple variables to stabilize production, optimize gas lift usage, and minimize human intervention across complex production environments. Key lessons on sensor validation, inferential modeling, legacy DCS integration, and step-test based model tuning have been carried forward and expanded upon in the present deployment of a cloud-connected, scalable gas lift optimization system.

Motivation for Study

The upstream oil and gas sector is undergoing a rapid digital transformation, driven by the convergence of artificial intelligence (AI), advanced analytics, and automation. In the domain of artificial lift, these technologies present a unique opportunity to transition from reactive, manual interventions to predictive, closed-loop optimization that can continuously adapt to changing reservoir and facility conditions.

However, translating this potential into field-scale value particularly in mature, brownfield environments remain complex. Legacy infrastructure often lacks the instrumentation density, data quality, and control system interoperability required for real-time optimization. Integrating AI driven solutions into such heterogeneous environments demands robust data engineering, secure and reliable communication layers, and scalable architectures that can coexist with existing control frameworks.

Furthermore, the operational shift from human-led decision-making to machine-assisted autonomy introduces challenges in workforce adoption, governance, and change management. Reliability, explainability, and cybersecurity become critical considerations to ensure trust in AI outputs and safeguard production integrity.

This study addresses these challenges through the deployment of a cloud-native, AI enabled, autonomous gas-lift optimization system within Abu Dhabi's digital oil field ecosystem. It focuses on overcoming technical integration barriers, aligning with operational workflows, and demonstrating measurable production gains, thereby providing a blueprint for scalable deployment across diverse asset portfolios.

Field and System Overview

Abu Dhabi Digital Oil Field Context

The case study is based on a mature onshore oil field in Abu Dhabi, consisting of over two hundred producing wells using gas-lift. The field presents significant operational complexity due to the heterogeneity of reservoir characteristics, well productivity, and artificial lift performance across the asset. Wells exhibit varying production profiles influenced by reservoir depletion, fluid properties, and historical intervention strategies, requiring continuous optimization to sustain production targets. The field is integrated into a centralized infrastructure and is managed through a Distributed Control System (DCS) that provides real-time monitoring and supervisory control. Despite this level of instrumentation, historical well management practices relied heavily on manual processes. Lift gas allocation was traditionally handled using scheduled, periodic well testing and operator-driven adjustments based on field experience, heuristic rules, and empirical understanding of well behavior. These legacy methods, while operationally feasible, often resulted in suboptimal gas utilization, delayed responses to dynamic reservoir conditions, and limited visibility into well performance between test cycles.

The increasing need for production efficiency, energy optimization, and reduced operational overhead provided a strong incentive to adopt a more automated and intelligent solution. This context provided the foundation for implementing the Well Autonomous Control System (AWC) a scalable, AI enabled autonomous control framework designed to modernize field operations, enhance well-level optimization, and improve overall lift gas allocation efficiency.

Existing Infrastructure and Limitations

The field utilized a conventional SCADA-DCS setup with minimal data analytics capability. Lift gas optimization decisions were decentralized and reactive, often lagging real-time well dynamics. System fragmentation, lack of integrated analytics, and aging hardware limited the efficiency of optimization efforts.

Architecture and Methodology

The proposed solution framework integrates both the operational technology (OT) network layers (Levels 1–3) and the enterprise business network layer (Level 4), enabling seamless and secure optimization across domains. At Level 3, a centralized APC performs multivariable predictive control on gas-lifted wells. It interfaces directly with the DCS, which supplies real time field data including pressures, temperatures, bottom-hole and wellhead conditions, flowline metrics, and valve positions. This data is processed by the APC server located in the field control room, where predictive models (described in following sections) analyze well behavior and compute optimal control actions to adjust choke valve actuators, ensuring reservoir integrity and adherence to operational constraints. Communication between the APC Server and wellsite controllers used secure, latency tolerant protocols over fiber optic network to maintain system stability and data integrity.

At Level 4, a cloud-hosted support tool complements the APC by monitoring performance trends, managing model updates in response to changing well and reservoir conditions, and guiding engineers in refining optimization strategies. This tool is integrated with corporate historian and petroleum engineering digital platforms, ensuring continuity and contextual awareness.

A secure data flow from field to Business network & cloud is maintained via a plant data historian and a unidirectional data diode, while PI Asset Framework (AF) structures well data for efficient access and analysis. This dual network architecture delivers a robust, scalable, and secure foundation for real-time, AI-driven gas-lift optimization.

Figures 1 and 2 provide a schematic overview of the system's architectural layers and the corresponding data flows that underpin the Autonomous Gas Lift Optimization System. Figure 1 illustrates the high-level system architecture, depicting the integration points between field instrumentation, centralized control elements, and cloud-enabled analytics. This layered architecture ensures that each domain from wellhead sensors and local control hardware to enterprise platforms and cloud resources contributes to a cohesive automation and optimization strategy. Figure 2 details the data pathways and repositories involved in real-time operations.

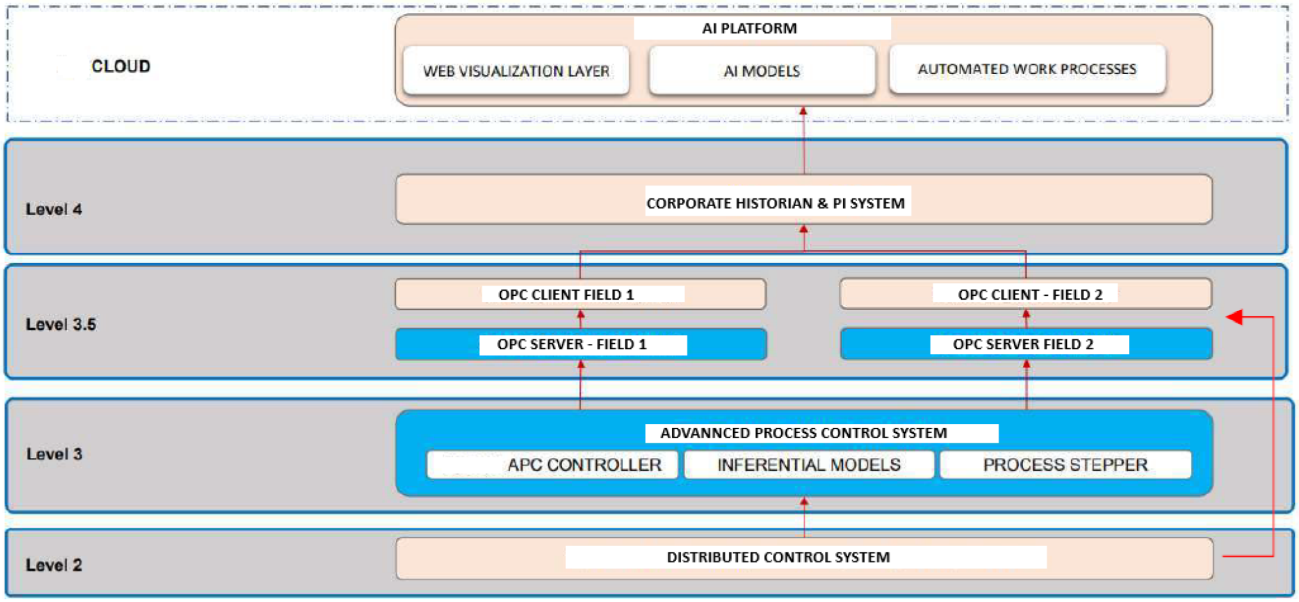

Figure 1—High Level System Architecture of Autonomous Gas Lift Optimization System

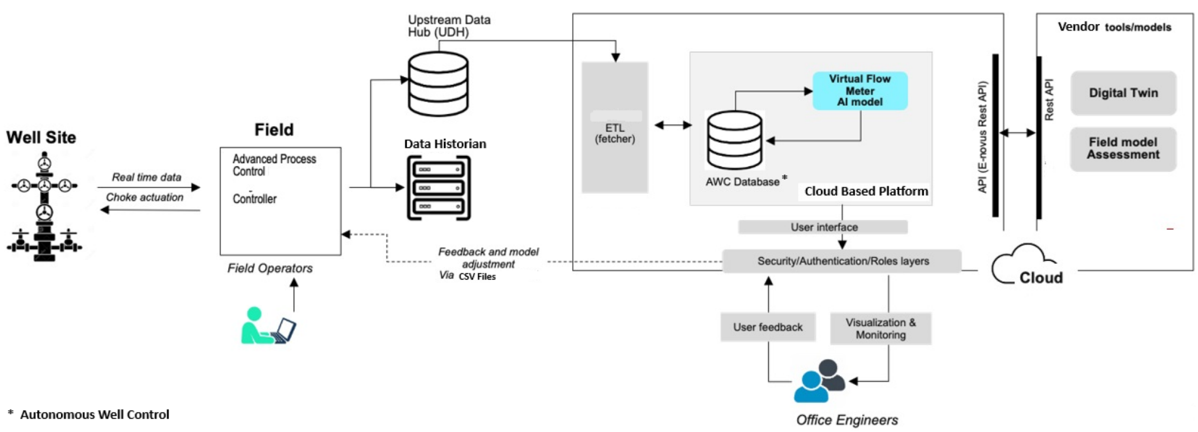

Figure 2—High Level Data Flow and Data Sources and Repositories

Hybrid Autonomous Control Framework for Gas Lift Well Optimization

The well autonomous control system (AWC) uses a hybrid framework combining AI modelling tools—Response Surface Methodology (RSM) and Artificial Neural Networks (ANNs)—to improve gas lift well modelling and real-time optimization.

RSM-based inferential models serve as surrogate representations of well performance, employing polynomial regression techniques to estimate key parameters such as bottomhole pressure and oil production rate. These estimations are derived from operational inputs including wellhead pressure, temperature, gas lift injection rate, gas-oil ratio (GOR), water cut, and productivity index. The surrogate models replicate the behaviour of calibrated well simulations with high fidelity, enabling efficient and transparent control logic suitable for field deployment. To maintain accuracy, model coefficients are periodically updated to reflect changes in well conditions and operational constraints. This approach supports high-frequency optimization without the computational overhead associated with full-physics simulations. Related work processes are described in the following sections.

In parallel, cloud-hosted ANNs function as Virtual Flow Meters (VFMs), learning complex production dynamics directly from real-time sensor data. ANN predictions are continuously cross validated against RSM outputs to ensure consistency and reliability. By combining explainable surrogate modelling with adaptive machine learning, this framework enables robust, real-time optimization of gas lift operations, improving production efficiency while maintaining operational transparency. For comprehensive details on the underlying modelling technologies and the digital operating model that enable this autonomous framework, readers are referred to Paper Number SPE-229402-MS.

Field based Work Processes

Field operational work processes in the Well Autonomous Control system are designed to enable fully autonomous well control while ensuring safety, reliability, and compliance with operational boundaries. These processes include continuous sensor validation routines to ensure the accuracy and consistency of real-time data, which is essential for reliable control performance. Multi-rate well testing is performed periodically to evaluate performance under varying conditions and to recalibrate model assumptions as needed. The APC dynamically adjusts the gas lift and production chokes in real time based on the inferential model outputs and optimization logic. Additionally, any manual interventions or system overrides are logged and time-stamped, supporting operational transparency and diagnostic tracking. Collectively, these field-level workflows ensure that the system remains robust, adaptive, and aligned with production goals while minimizing the need for manual oversight.

Business Network & Cloud-based Work Processes

Building on the foundational implementation presented in Hernandez et al. (2024), where inferential modeling, bias correction, and parameter update methodologies were established to support autonomous well control at the field level, this work extends those concepts into a fully cloud native architecture. The transition enables scalable, automated lifecycle management of inferential models across multi-well environments, enhancing system reliability, adaptability, and control responsiveness.

In this evolution, a suite of cloud hosted workflows has been developed to maintain model fidelity and ensure that the Autonomous Control System adapts in real time to changing conditions. This shift from static, engineer-led calibration to dynamic, data-driven processes aligns with modern digital oilfield strategies. Leveraging scalable cloud infrastructure, the system supports asynchronous, field-wide model updates across hundreds of wells minimizing engineering effort, improving operational continuity, and enabling sustained optimization at scale.

Cloud-based workflows can be categorized into three key components:

Inferential Model Generation Workflow. Inferential models are constructed to estimate key variables (e.g., flowing bottomhole pressure, oil production rate) that are either not directly measured or prone to sensor degradation. The model generation workflow executes entirely within the cloud and is designed to operate autonomously based on real-time triggers and data streams.

Workflow Capabilities:

- Generation of inferential coefficients through regression against synthetic and historical data.
- Continuous validation against performance thresholds, with dynamic flagging of models exhibiting calibration drift or input anomalies.
- This workflow enables continuous model adaptation and ensures inferential models remain consistent with evolving well behaviour and operational states.
- This workflow orchestrates the creation and update of inferential models based on real-time operational data and predefined source well models. It is triggered either on a scheduled basis or in response to detected changes in well behaviour.

Bias Correction Workflow. Post-deployment, inferential models are continuously assessed for alignment with actual well performance. The bias correction workflow performs a reconciliation between the model predictions and available field measurements to maintain predictive accuracy over time.

Workflow Capabilities:

- Compares inferential outputs (e.g., calculated oil rate) with measured well parameters from validated flow test data.
- Calculates the bias and applies corrections to the inferential model where discrepancies exceed a configurable threshold.
- Updates the model coefficients or applies offset terms to maintain accuracy.
- Flags models for recalibration if bias persists over multiple evaluation windows, helping to avoid control decisions based on degraded predictions.

Bulk Parameter Update. The Bulk Parameter Update workflow facilitates the secure and efficient propagation of updated inferential model coefficients, bias correction terms, and operational boundaries to field-deployed Advanced Process Control (APC) systems. By aggregating these updates into structured, version-controlled files (e.g., CSV or JSON formats), the workflow ensures consistent and traceable synchronization across the control network. This capability enables the rapid deployment of model enhancements and control logic modifications, allowing the system to adapt swiftly to evolving reservoir conditions, equipment constraints, or facility operating limits all while maintaining operational continuity and system stability.

The figure below illustrates the architecture and interaction of the cloud hosted workflows that govern the lifecycle of inferential models within the autonomous gas lift optimization system. It captures the end to end flow from model generation and bias correction to bulk parameter updates highlighting how each workflow component contributes to maintaining model accuracy, enabling real-time control decisions, and ensuring seamless synchronization with field-deployed APC systems.

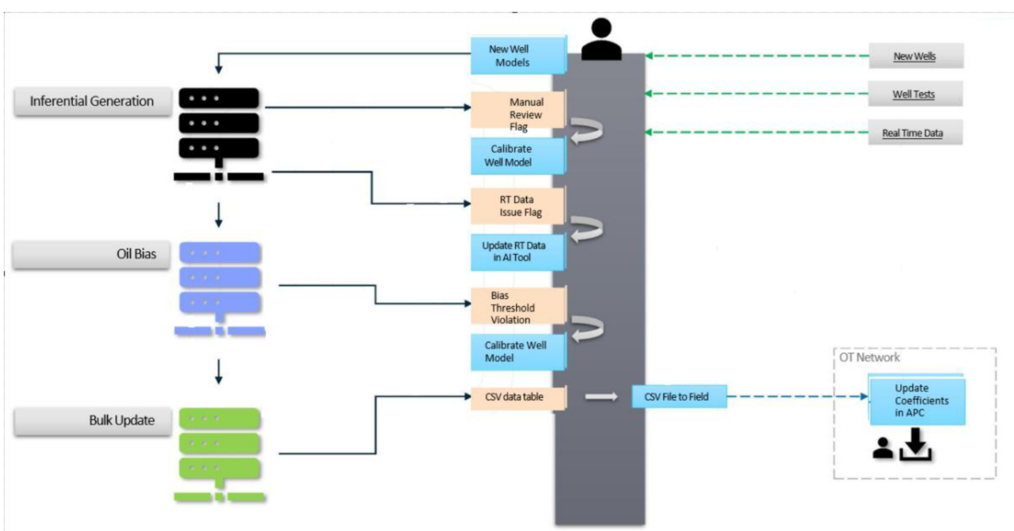

Figure 3—Business Network Work processes for New Model generation and updating existing Models

Cross-Functional Collaboration and System Integration

The successful deployment of AI enabled autonomous gas lift optimization was made possible through close collaboration among cross-functional teams, comprising petroleum engineers, data scientists, IT/OT specialists, and control engineers. These teams jointly developed, tested, and validated the end-to-end optimization workflows, ensuring technical robustness and operational reliability. Legacy system compatibility was addressed using middleware connectors and standardized interfaces, enabling seamless integration with existing infrastructure. To safeguard sensitive operational data, comprehensive cybersecurity measures were implemented in alignment with corporate standards, ensuring both data sovereignty and regulatory compliance.

In line with best practices adopted by major International Oil Companies and several National Oil Companies, the project team also emphasized the importance of integrating Allocation & Assurance, Metering, and Real-Time Data Management specialists within the core team. This integrated approach ensured high-quality, validated data was consistently available to support reliable and secure autonomous operations.

Challenges and Mitigation Strategies

The implementation of AI enabled autonomous gas-lift optimization in digital oil fields introduces several technical and operational challenges. Below is a detailed breakdown of each challenge along with proposed strategies to overcome them:

Challenge: Geographically Scattered Wells and Control Coordination

One of the key challenges in implementing autonomous gas lift optimization was managing data acquisition and control coordination across a vast number of oil-producing wells distributed over large geographic areas. This spatial dispersion introduced significant complexity in ensuring timely, reliable data transmission and effective centralized decision-making, particularly given the constraints of existing infrastructure and the need for real-time responsiveness.

To mitigate this, a hierarchical control architecture was deployed, integrating field-level computing capabilities at the central processing plant with existing Distributed Control System (DCS) controllers at remote wellsite locations. This configuration enables local preprocessing of field data at the APC and DCS layers, transmitting only essential parameters to cloud-based enterprise servers. The result is a substantial reduction in bandwidth usage and latency, facilitating efficient centralized decision-making.

The solution leverages existing control network infrastructure, preferably based on fibre-optic communication and robust industrial protocols such as OPC, to ensure secure and resilient connectivity. This architecture provides a scalable foundation for coordinating autonomous well control across distributed field environments, supporting real-time optimization while minimizing operational overhead.

Challenge: Variability in Choke Trims and Flow Control Standardization

A significant challenge in field-wide gas lift optimization was the variability in gas lift and production choke trims installed across different producer wells. The actuator reported choke travel percentage often lacks a consistent correlation with actual flow rate due to differences in trim geometry and valve behaviour, making it unsuitable as a standard control parameter across the field. This inconsistency introduces nonlinearities and control inaccuracies, particularly in centralized APC implementations.

To address this, the team implemented a choke transformation strategy based on equivalent choke bean size, which provides a normalized representation of choke position independent of hardware variability. This transformation was integrated into the APC logic, enabling consistent control behaviour across wells. Additionally, model parameterization using well-specific productivity index (PI) and choke flow coefficients allowed the system to better capture unique well responses and predict flow behaviour more accurately. To further enhance control reliability, adaptive control algorithms were introduced to dynamically learn and adjust choke response over time, ensuring robust performance despite hardware and reservoir heterogeneity.

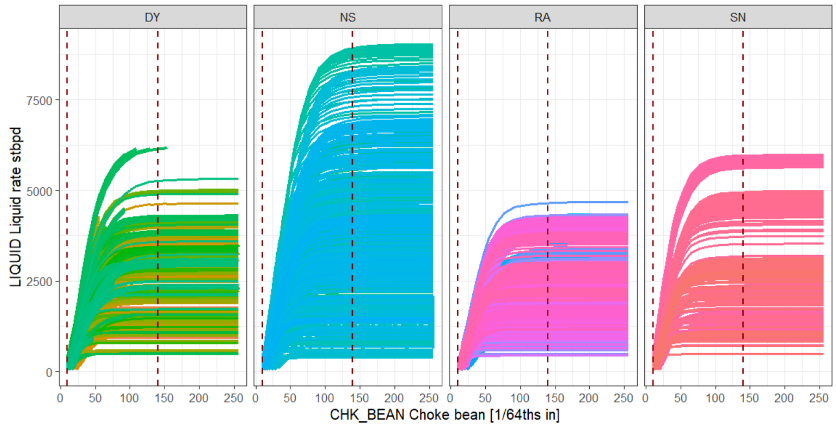

Figure 4—Choke Bean vs. Liquid rates curves for 200+ wells & different gas lift rates

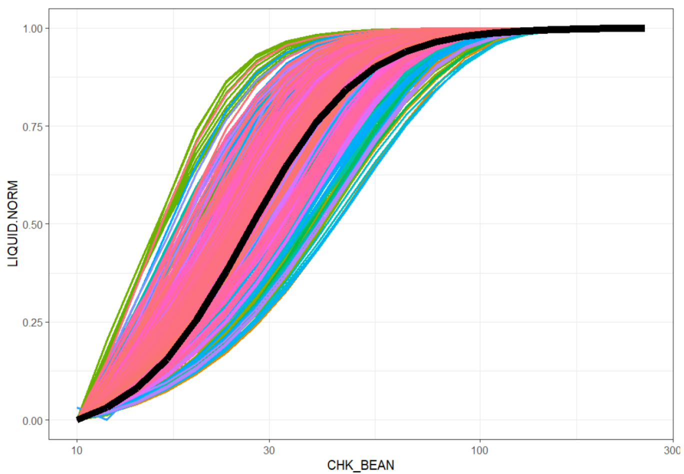

Figure 5—Choked Bean vs. Normalized Liquid

Challenge: Unavailability and Intermittency of Critical Well Measurements

A recurring challenge in well autonomous control system (AWC) is the absence of continuous measurements for key parameters such as bottomhole pressure (BHP) and the oil flow rate data. In the studied field, bottomhole pressure is available for only 15% of the wells, while real-time oil rate measurements are not continuously captured. Instead, wells are periodically routed to a multiphase flow meter (MPFM) for testing, which provides only intermittent validation points. These data gaps often stemming from instrumentation limitations or the high cost of permanent sensors can compromise real-time optimization and model fidelity.

To address this challenge, the solution implemented inferential modelling approaches that leverage surrogate variables including wellhead pressure, temperature, and valve positions to estimate unmeasured parameters in real time. For flow rate estimation, virtual flow metering (VFM) techniques were deployed and periodically validated against well test data to ensure ongoing accuracy. BHP was estimated using physics-informed inferential models, calibrated with historical production trends and reservoir simulation outputs.

Furthermore, uncertainty quantification was integrated into the modelling framework to support robust control decisions under ambiguous or partially observed conditions. This multi-layered approach ensured reliable autonomous operation even in wells with limited direct instrumentation.

Challenge: Low-Fidelity Well Models Due to Limited Data Availability

Developing accurate dynamic models for gas lift optimization is often hindered by limited historical data, missing measurements, and incomplete well testing records, particularly in brownfield environments. These constraints result in low-fidelity well models, reducing the effectiveness of model-based control strategies and limiting the reliability of production forecasts.

To overcome this challenge, the solution implemented an AI-driven recalibration framework, which continuously refines well models based on field testing data and real-time performance feedback. Where direct data was unavailable, synthetic data generation from reservoir simulations and analogue wells was used to augment the model training sets. Additionally, transfer learning techniques were applied to leverage operational insights and model parameters from geologically and operationally similar wells or fields, significantly improving initial model accuracy and reducing calibration time.

This multi-pronged approach enabled the development of robust, adaptive well models, capable of supporting reliable control even in data scarce conditions ultimately enhancing the performance of the autonomous gas lift optimization system across a diverse well population.

Challenge: Sensor and Valve Failure in Autonomous Operations

Ensuring reliability in well autonomous control system (AWC) for gas lift optimization is particularly challenging due to the potential for sensor malfunctions and valve failures, which can compromise both safety and optimization performance. Faulty limit switches, inconsistent valve feedback, and intermittent sensor data can result in unintended control actions, data integrity issues, or even unsafe operating conditions in the absence of proper safeguards.

To mitigate these risks, a robust watchdog & shed logic framework was implemented within the APC & DCS system. This logic prioritizes command-based valve control over feedback signals to avoid misinterpretation caused by faulty limit switches. In the event that critical safety valves (e.g., SSSV, GL SDV, MSASV) are closed or operating outside expected limits, the system automatically drops the associated control setpoints and safely disables the controller, preventing unsafe actuation. Furthermore, anomalous sensor readings such as frozen or out-of-range values for BHP or casing pressure trigger safety interlocks, suspending optimization until reliable data is restored.

The system also includes automated diagnostics and fail-safes, such as disabling manual valve activation or APC controller engagement when setpoints exceed predefined safe limits (such as Rate of change for process variables). These measures ensure that autonomous operation can continue safely, while minimizing manual intervention and maintaining overall system integrity in the face of equipment failure.

Table 1—APC Controller scenarios where Optimization is disabled

Scenario #	Operator CVs ON					
	OIL	PWF	BHP_A	WHP	Condition	Action
0	Any				Well On Test	Disable Optimization (MV does not move, unless any active CV limit is violated)
1	0	0	0	1	Calculation or Signal Conditioning Error on selected CVs	
2	0	0	1	0		
3	0	0	1	1		
4	0	1	0	0		
5	0	1	0	1		
6	0	1	1	0		
7	0	1	1	1		
8	1	0	0	0		
9	1	0	0	1		
10	1	0	1	0		
11	1	0	1	1		
12	1	1	0	0		
13	1	1	0	1		
14	1	1	1	0		
15	1	1	1	1		

Challenge: Model Updates in OT Environments with Security Constraints

A key operational challenge in deploying AI-driven gas lift optimization lies in updating models within the OT (Operational Technology) environment, where strict cybersecurity protocols limit bidirectional communication between the cloud-based analytics layer (Level 4 business network) and the on-site APC server (Level 3 control network). These restrictions are critical for protecting operational assets but create barriers to real-time model synchronization, impacting the agility of adaptive optimization workflows.

To mitigate this constraint, the solution adopted a secure, offline model update mechanism. Model coefficients, target rates, and relevant parameters were periodically exported from the cloud system as CSV files, and then manually transferred and uploaded to the Level 3 APC server using approved secure transfer protocols during designated update windows. Additionally, models were containerized to ensure compatibility and portability, allowing them to be deployed offline and synchronized during planned maintenance periods without exposing the control network to cyber risks.

This approach balances the need for model adaptability and security compliance, ensuring that the optimization system remains updated and effective, while maintaining strict separation between IT and OT layers in line with industrial cybersecurity standards.

Observations & Results

The deployment of Autonomous Control systems in digital oil fields yielded several quantifiable and qualitative outcomes, validating the effectiveness of the proposed mitigation strategies across multiple operational challenges.

Geographically Scattered Wells and Control Coordination

The deployment of hierarchical control architecture significantly improved coordination across geographically dispersed wells. By integrating field-level computing at the central processing plant with existing Distributed Control System (DCS) controllers at remote well sites, the system enabled localized preprocessing of operational data.

This architecture facilitated real-time centralized decision-making without overloading the communication infrastructure. Detailed sizing of existing DCS/SCADA points enabled efficient allocation and reuse of infrastructure, supporting the scale-up without additional hardware. The use of fibre-optic networks and industrial protocols such as OPC ensured secure and resilient connectivity. Field trials demonstrated that autonomous control actions were executed within acceptable latency thresholds, even in remote locations, validating the scalability and robustness of the solution.

Variability in Choke Trims and Flow Control Standardization

The implementation of a choke transformation strategy based on equivalent choke bean size led to a marked improvement in control consistency across wells. By normalizing choke position data, the system eliminated discrepancies caused by hardware variability.

Further enhancements were achieved through well-specific model parameterization using productivity index (PI) and choke flow coefficients. Adaptive control algorithms dynamically adjusted to evolving well conditions, maintaining optimal performance. Field validation showed that the system sustained stable gas lift operations across wells with diverse choke configurations, confirming the effectiveness of the standardization approach. Furthermore, oil production targets were achieved with marginal error below 20 BLS difference, demonstrating that the effectiveness of standardization approach across multiple wells. Refer to Figure 6 below for an example demonstrating how AWC enabled the achievement of the oil target precisely. However, high GOR wells required further extension of the model at low opening.

Figure 6—Optimized Choke Transformation and Control Standardization for Consistent Oil Production Across Wells

Unavailability and Intermittency of Critical Well Measurements

The inferential modelling framework enabled reliable autonomous gas lift optimization across wells with limited instrumentation. Real-time estimation of bottomhole flowing pressure (BHFP) and flow rates is a game changer in the process of production optimization facilitating well surveillance and operational decision making, especially in fields without downhole pressure gauges. The inferred values showed strong alignment with periodic well test data, supporting high model fidelity. Figure 7 illustrates the fidelity of the Production Rate Inferential model (PRIM), as validated against calibrated well model. The accuracy levels depend on the quality and consistency of the well tests used to calibrate the well models. For good quality models the errors are as small as 0.5psi as shown in Figure 7.

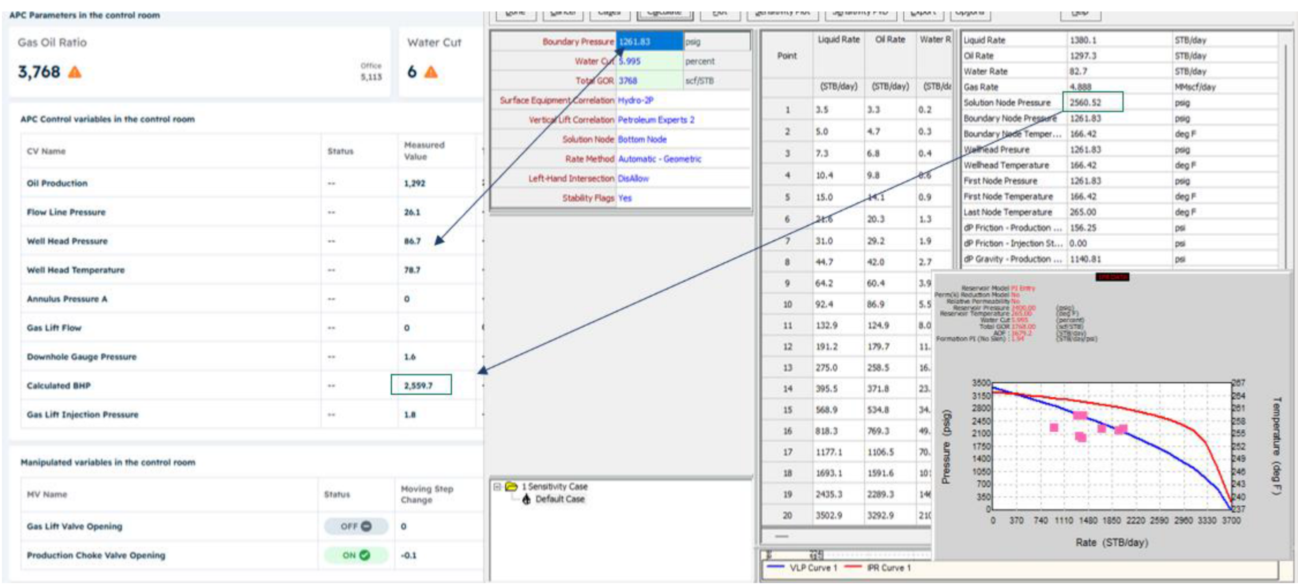

Figure 7—Validation of Production Rate Inferential Model Against Calibrated Well model

Below Figure. 8 illustrates the fidelity of the Machine learning (ML) virtual flow metering (VFM) model, as validated against multiphase flow meter (MPFM) well test data. The comparison yielded a mean absolute percentage error (MAPE) of approximately 4.62%, indicating strong agreement between the model predictions and measured values.

Figure 8—Validation of Machine Learning Virtual Flow Metering Model Against Multiphase Flow Meter (MPFM)

Notably, the accuracy of BHFP predictions for gas lifted wells was significantly influenced by the quality of the measurements, recent pressure surveys, and representative well tests. Under optimal conditions, BHFP prediction errors were typically constrained within a narrow range of 1–200 psi. Figure 9 illustrates the fidelity of the Bottom Hole Flowing Pressure (BHFP) inferential model as validated against downhole pressure gauge.

Figure 9—Validation of Bottom Hole Flowing Pressure (BHFP) Model Against Downhole Pressure Gauge

Sensor and Valve Failure in Autonomous Operations

The implementation of the watchdog and shed logic framework substantially improved the reliability and safety of autonomous gas lift operations. Wells previously prone to control interruptions due to faulty limit switches, inconsistent valve feedback, or anomalous sensor readings were able to maintain stable performance under advanced process control (APC). The system consistently prevented unsafe actuation by automatically disengaging APC controllers when critical valves operated outside expected limits or when sensor data became unreliable.

As a result, the number of wells operating under APC increased, with extended runtime and reduced manual intervention. Wells remained within safe operating envelopes, and optimization continuity was preserved even in the presence of equipment failures. This translated into enhanced system integrity, improved operational uptime, and greater confidence in autonomous field-wide optimization.

Figure 10—Enhanced Solution Uptime with Watchdog & Shed Logic Strategy Implementation

Model Updates in OT Environments with Security Constraints

The secure offline model update process fully complied with cybersecurity protocols and internal policy requirements, ensuring zero exposure to operational risk. By enabling bulk updates of limits, targets, and model parameters, the approach significantly enhanced operational efficiency. Compared to traditional well-by-well manual updates, the new method delivered a time savings of approximately 98%, allowing for faster deployment and synchronization of recalibrated optimization models. This efficiency translated into improved system responsiveness and sustained productivity across the gas lift network.

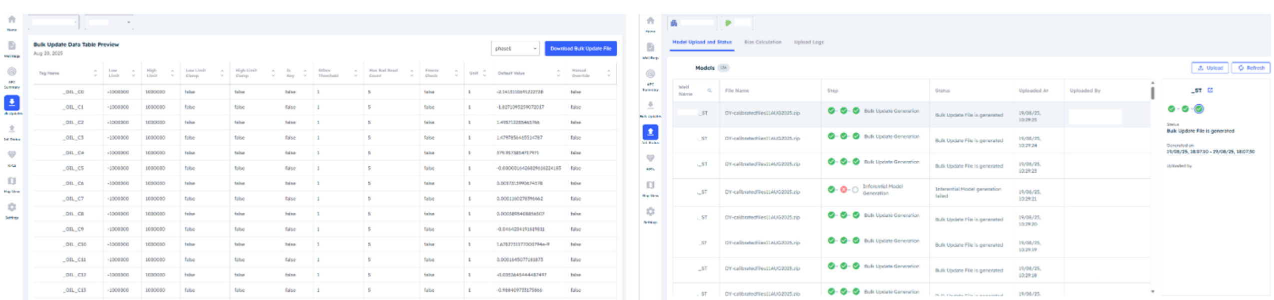

Figure 11—Bulk Update Data Table Preview and Model Upload Status Interface

Production Uplift & Lift Gas Optimisation

Prior to the implementation of autonomous control, the gas-lifted well was managed manually, with operators responsible for initiating gas injection and production. Upon transitioning to autonomous operations, a notable enhancement in well performance was observed. Figure 12 shows a well example with an oil production increased approximately by 7% (MPFM), while gas lift injection rates were optimized from 2.8 MMSCFD to 2.3 MMSCFD, representing an improvement of nearly 21% in gas utilization efficiency. The periods of active Well Autonomous Control System (AWC) operation are denoted in green, and intervals of manual control (AWC inactive) are highlighted in light blue while MPFM test results in yellow. These results demonstrate a clear shift in operational effectiveness attributable to autonomous optimization.

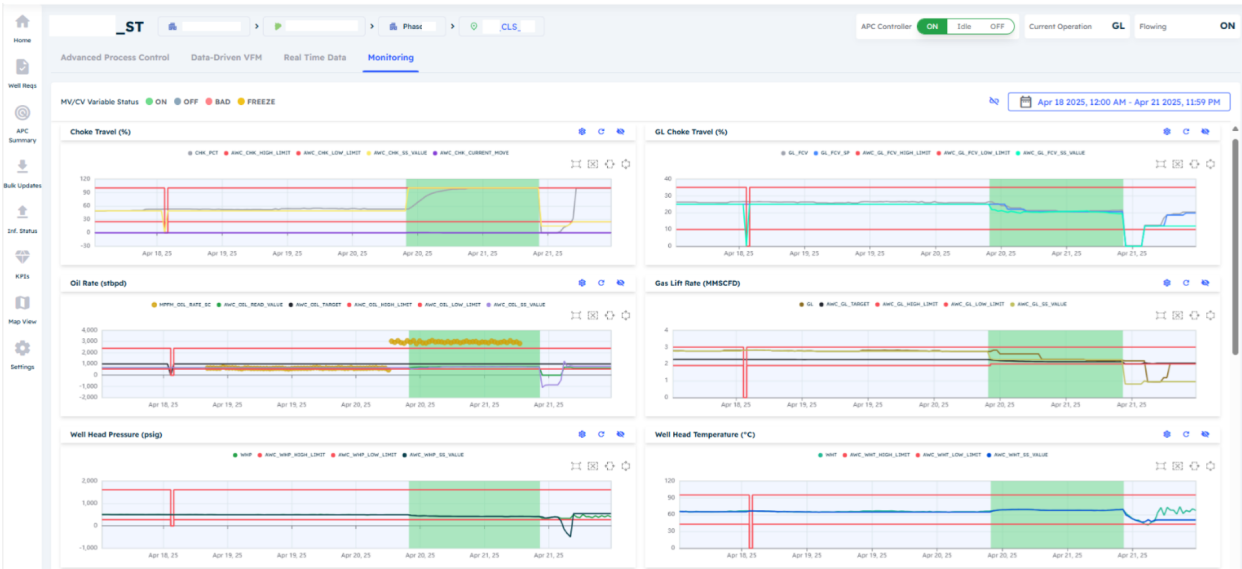

Figure 12—Impact of AWC on Well Performance Metrics

Bottom Hole Pressure Compliance

Another critical advantage of autonomous control lies in ensuring strict adherence to operational and reservoir management protocols, particularly the compliance of bottomhole pressure within predefined upper and lower limits. In addition, maintaining compliance with oil production targets, gas–oil ratio (GOR) thresholds, and facility design constraints for temperature and pressure is essential. Advanced Process Control (APC) systems facilitate this by embedding high and low setpoints for key process variables directly into the controller logic to align with field operating requirements. The implementation of such technology enables consistent regulatory compliance and optimizes energy efficiency across the production lifecycle. Below Figure 13. shows bottomhole pressure lower limit is systematically honoured & enforced to safeguard reservoir integrity and operational reliability of a sample well.

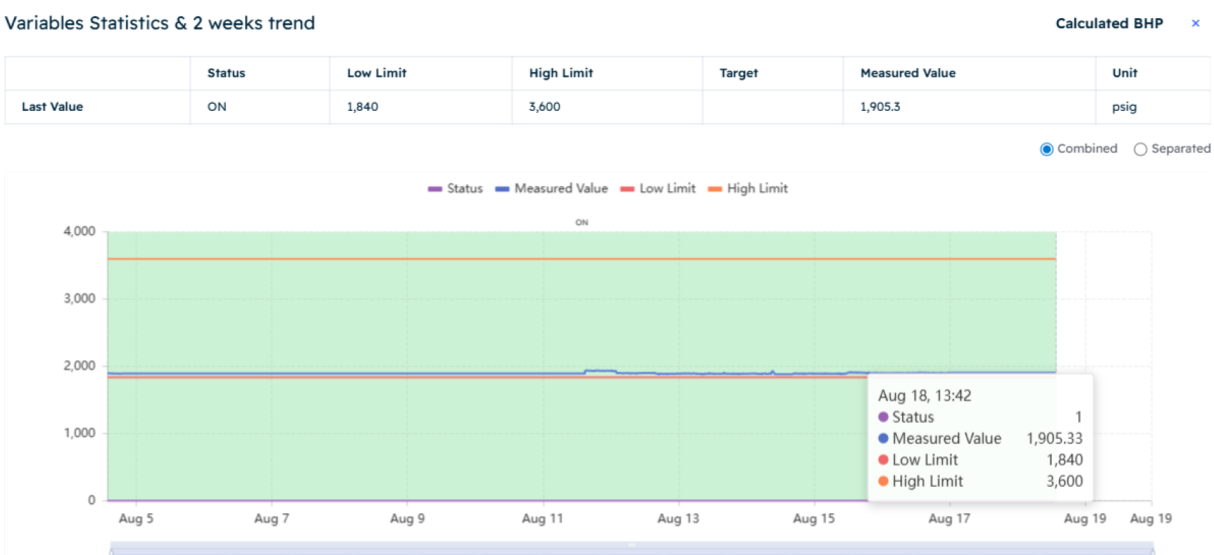

Figure 13—Compliance and Control of Bottomhole Pressure Under Autonomous Operation

Below Figure 14. shows compliance of operational requirements and constraints of a well. Left spider chart indicates operating ranges programmed within the APC controller while right spider chart indicates measured values for key process variables complying operating ranges.

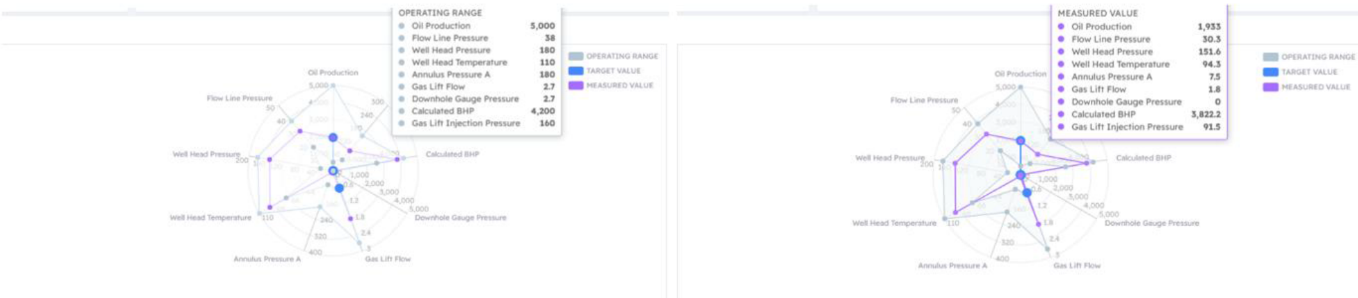

Figure 14—AWC Performance: Achieving Operational Standards and Compliance

Summary of Quantitative KPIs observed post-implementation:

The above results in Table 2, underscore the robustness of the hybrid modelling and control framework, particularly in addressing challenges such as sensor intermittency, choke variability, and geographically distributed well coordination. The system's ability to maintain high uptime, reduce manual workload, and operate safely under uncertain conditions reflects its maturity and readiness for broader field deployment.

Table 2—AWC Performance Metrics Summary

Performance Metrics	Improvement Achieved
Production Uplift	Up to 1-2% increase production across deployed wells
Gas Lift Savings	Up to 5% -30% optimisation at well level, across deployed wells
Bottom Hole Pressure Compliance (+/-50 Psi)	100%
Manual intervention	50% reduction
Flow Prediction accuracy	+/-10% (VFM vs MPFM)
Safety Incident Rate	0% during autonomous operation

Conclusions

The Abu Dhabi Digital Oil Field case study validates the feasibility and effectiveness of AI-enabled autonomous gas lift optimization in brownfield environments. While operational and technical challenges were present, these were successfully addressed through a combination of advanced modelling techniques, secure system integration, and strong cross-disciplinary collaboration. The solution delivered measurable gains in production efficiency and system responsiveness, underscoring the value of AI in mature asset operations.

This paper presents a field-proven methodology for Well Autonomous Control System (AWC) based on a centralized Advanced Process Control (APC) system integrated with cloud-based analytics. Unlike traditional distributed edge computing, this centralized architecture simplifies deployment, reduces hardware footprint, and enhances system manageability. The approach is broadly applicable to assets with similar artificial lift systems and legacy infrastructure constraints.

The success of this deployment enabled by cross-functional collaboration and intelligent automation offers a replicable blueprint for digital transformation in upstream operations. This work lays the foundation for broader adoption of AI-driven optimization across diverse production environments.

Recommendations and Future Work

Future work should focus on scaling AI-based optimization from individual wells to well clusters or field-level deployments, enabling intelligent coordination of gas lift across multiple wells sharing common infrastructure or constraints. This approach can unlock additional gains in gas utilization efficiency, lift gas distribution, and overall field productivity.

Expansion beyond gas lift to include water injection and ESP wells will further support a holistic, integrated strategy for field-wide production optimization. To build greater operator trust, future systems must enhance the interpretability of AI models, providing transparent justifications for control actions.

In parallel, federated learning should be explored to enable cross-asset intelligence while maintaining data privacy and eliminating the need for centralized data aggregation. Additionally, reinforcing AI governance frameworks and ensuring auditability will be essential to meet operational and regulatory standards.

Finally, the establishment of human-in-the-loop systems remains critical for validating and supervising AI-driven control in safety-critical environments, ensuring a balance between autonomy and oversight in next-generation production operations.

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Nomenclature

AI : Artificial Intelligence
ANN : Artificial Neural Network
APC : Advanced Process Control
AWC : Well Autonomous Control System
BHFP : Bottom Hole Flowing Pressure
CSV : Comma Separated Values
CV : Controlled Variable
DCS : Distributed Control System
ETL : Extract, Transform, Load
GL : Gas Lift
GOR : Gas Oil Ratio
IT : Information technology
JSON : JavaScript Object Notation
KPI : Key Performance Indicator
MAPE : Mean Absolute Percentage Error
ML : Machine Learning
MPC : Multivariable Predictive Control
MPFM : Multi- Phase Flow Meter
MSASV : Master Surface Annulus Safety Valve
MV : Manipulated Variables
OPC : OLE for Process Control
OT : Operator Technology
PI : Productivity Index

PRIM : Production Rate Inferential Model
PWF : Flowing Bottom Hole Pressure
RSM : Response Surface Methodology
SCADA : Supervisory Control and Data Acquisition
SDV : Shut-Down Valve
SSSV : Sub Surface Safety Valve
VFM : Virtual Flow Meter
WC : Water Cut
WHP : Well Head Pressure

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